Lecture 06, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

Theorem of Existence and Uniqueness for 1st Order IVP (cont.)

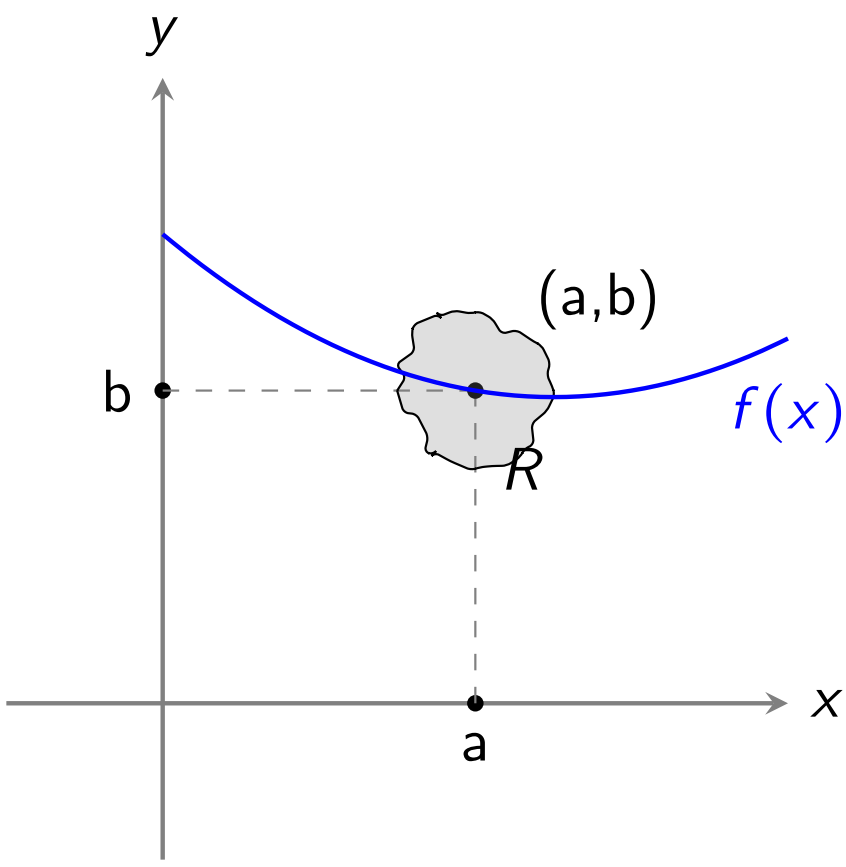
From last lecture: given the 1st order IVP

$$\begin{cases} y' = f(x, y) & \text{(where } f \text{ may be nonlinear)} \\ y(a) = b \end{cases}$$

Theorem: If $\exists$ a simply connected region R in the x, y plane that includes (a, b) and if $f(x, y)$ is continuous in R then $\exists$ a solution in an interval around $x = a$. And if $\frac{\partial f}{\partial y}(x, y)$ is also continuous in R then the solution is unique.

timestamp: 00:00

Theorem of Existence and Uniqueness for 1st Order IVP (cont.)



If $f(x)$ is continuous in the shaded region R, then $\exists$ a solution in an interval around $x = a$. And if $\frac{\partial f}{\partial y}(x, y)$ is also continuous in R, then the solution is unique.

timestamp: 02:12

Ex: consider the equation $\frac{\mathrm{d}y}{\mathrm{d}x} = 2\,x\,y^2$.

Here $f(x, y) = 2\,x\,y^2$ and $\frac{\partial f}{\partial y}(x, y) = 4\,x\,y$.

Both f and $\frac{\partial f}{\partial y}$ are continuous everywhere, thus for any given initial condition we can find a region R around it where f and $\frac{\partial f}{\partial y}$ are continuous. According to the theorem just developed, we are guaranteed exactly one unique solution to the IVP in that region R.

Solving the IVP with the initial condition $y(0) = 1$

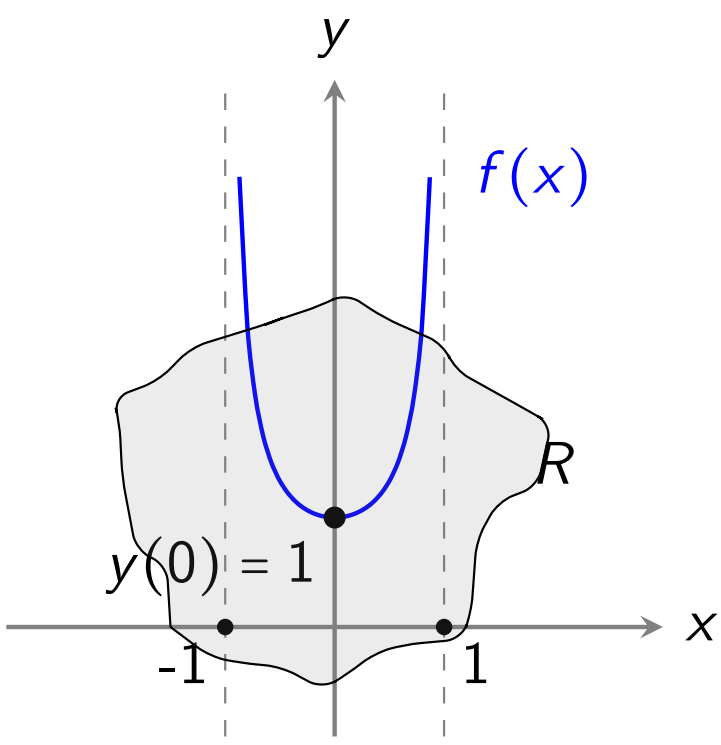
The equation is separable:

$$\frac{\mathrm{d}y}{y^2} = 2x\,\mathrm{d}x \;\rightarrow\; -\frac{1}{y} = x^2 + A \;\rightarrow\; y(x) = \frac{1}{A - x^2} \text{ (general solution)}$$

Applying the IC: $y(0) = \frac{1}{A} = 1$, so $A = 1$ and **the** solution to the initial value problem (around the initial condition) is

$$y(x) = \frac{1}{1 - x^2}.$$

timestamp: 04:35



Even though the region R where f and $\frac{\partial f}{\partial y}$ are continuous can be made arbitrarily large, this solution is valid only in the interval where $x \in (-1, 1)$. This theorem cannot predict the range where a solution exists—to figure that out we need to find the actual solution.

timestamp: 07:40

Ex:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2y}{x}$$

Here $f(x, y) = \frac{2y}{x}$ and $\frac{\partial f}{\partial y}(x, y) = \frac{2}{x}$

These equations are continuous everywhere except at $x = 0$. So if the initial condition is given somewhere at a $x \neq 0$, then the theorem guarantees a local unique solution.

timestamp: 11:14

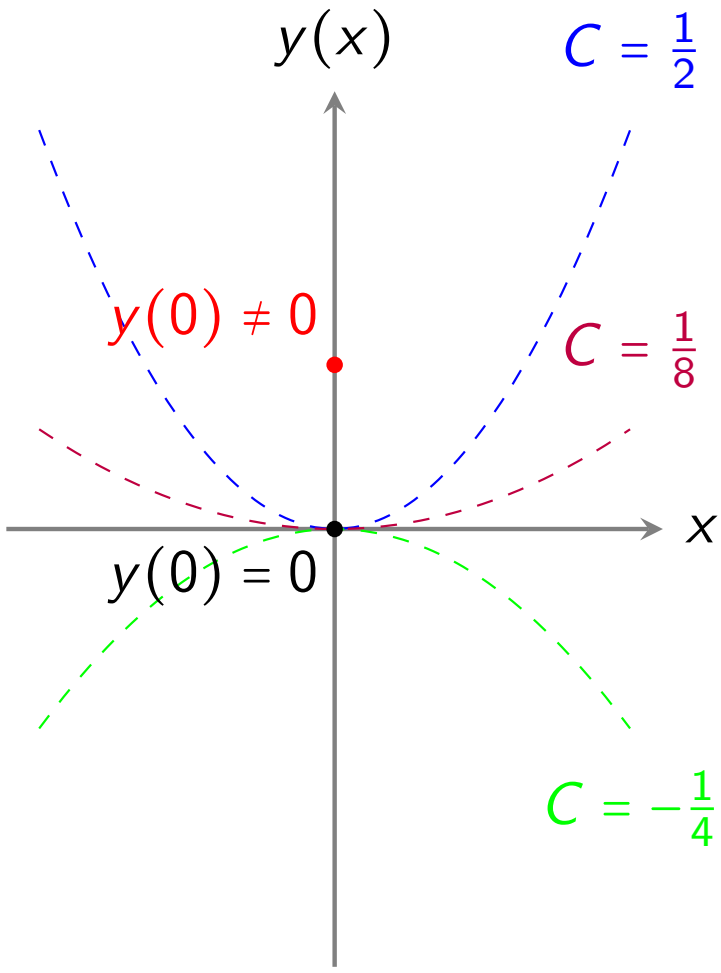
Solving this separable equation:

$$\frac{\mathrm{d}y}{y} = \frac{2\,\mathrm{d}x}{x} \;\rightarrow\; \ln(y) = 2\ln(x) + A \;\rightarrow\; y(x) = C\,x^2$$

Suppose the IC is $y(-1) = 1$. Then we get $y(-1) = C = 1$ and the solution is

$$y(x) = x^2$$

timestamp: 12:35



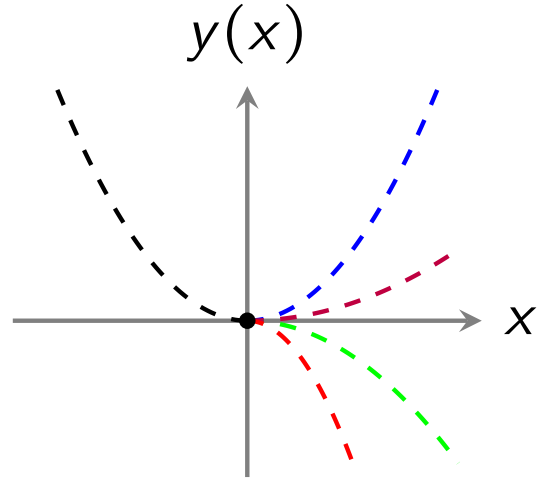
If the IC is $y(0) \neq 0$, then there is no solution, because no C would work.

If the IC is $y(0) = 0$ then there are infinite solutions, because any C would work. This does not contradict the theorem, because the theorem cannot be applied at $x = 0$, as neither $f(x, y)$ nor $\frac{\partial f}{\partial y}(x, y)$ are continuous around $x = 0$.

timestamp: 13:40

Note: even in the case the IC is $y(-1) = 1$ we can construct the following solutions

$$y(x) = \begin{cases} x^2 & \text{for } x \leq 0 \\ Cx^2 & \text{for } x > 0 \end{cases}$$



The solution is unique only up to $x = 0$, after that it is not. Again this is not inconsistent with the theorem, because the theorem only guarantees existence and uniqueness locally, i.e. somewhere around the initial condition.

timestamp: 16:00

Application: Water Mixing in a Tank

R: rate of water coming in = rate of water going out.

V: volume of water in the tank, fixed in time.

$C(t)$: concentration of salt in water in the tank.

C_{in}: concentration of salt in water coming in, fixed in time.

C_{out}: conc. of salt in water going out (= $C(t)$, perfect mixing)

$Q(t) = C(t) * V$: quantity of salt in water in tank.

timestamp: 22:00

The rate of change for $Q(t)$ is $\frac{\mathrm{d}Q}{\mathrm{d}t} = R \cdot C_{in} - R \cdot C_{out}$ which gives us the equation $\frac{\mathrm{d}C}{\mathrm{d}t} = \gamma\,(C_{in} - C)$ with $\gamma = \frac{R}{V}$. This is separable:

$$\frac{\mathrm{d}C}{C - C_{in}} = -\gamma\,\mathrm{d}t \;\rightarrow\; \ln(C - C_{in}) = -\gamma t + A$$

So the general solution is

$$C(t) = C_{in} + D\,\mathrm{e}^{-\gamma t}$$

timestamp: 25:05

If the IC is $C(0) = C_0$ then we get

$$C(0) = C_{in} + D = C_0 \;\rightarrow\; D = C_0 - C_{in}$$

and the solution is

$$C(t) = C_{in} + (C_0 - C_{in})\mathrm{e}^{-\gamma t}$$

Note: the concentration in the tank will tend to the C_{in} as time goes by.

timestamp: 29:25

Application: Emptying Tank

A: horizontal section of cylindrical tank.

$h(t)$: height of water in the tank.

a: section of hole at the bottom of the tank.

α: proportion of the hole that water comes out of $(0 < \alpha \leq 1)$

$v(t)$: velocity of water coming out of the hole (according to

Torricelli's law $v = \sqrt{2gh}$; g: local gravitational acceleration)

$A * h$: amount of water in the tank.

timestamp: 33:00

Rate of change of water in the tank:

$$\frac{\mathrm{d}Ah}{\mathrm{d}t} = -a\alpha v = -a\alpha\sqrt{2gh} \;\rightarrow\; \frac{\mathrm{d}h}{\mathrm{d}t} = -\gamma\sqrt{h}$$

where $\gamma = \frac{a\alpha\sqrt{2g}}{A}$

timestamp: 34:40

This is a separable equation:

$$\frac{\mathrm{d}h}{\sqrt{h}} = -\gamma\,\mathrm{d}t \;\rightarrow\; 2\sqrt{h} = -\gamma t + C \;\rightarrow\; \sqrt{h} = C - \frac{\gamma t}{2}$$

Applying an IC $h(0) = h_0$:

$$\sqrt{h(0)} = C = \sqrt{h_0}$$

and therefore

$$h(t) = \left(\sqrt{h_0} - \frac{\gamma t}{2}\right)^2$$

timestamp: 37:30